

Algebra 1
Unit 5
Lesson 1 Practice

Name _____
Date _____
Period _____

1. When solving an equation, the goal is to _____ the variable.

Two students, Josephine and Jackie, were solving the equation $39 - 8n = -8(3 + 4n) + 3n$. Consider their work shown below for questions 2–5.

Josephine's Work	Jackie's Work
$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-3n} \qquad \qquad \underline{-3n} \\ 39 - 11n = -8(3 + 4n) \\ 39 - 11n = -24 - 32n \\ \underline{+32n} \qquad \qquad \underline{+32n} \\ 39 + 21n = -24 \\ \underline{-39} \qquad \qquad \underline{-39} \\ 21n = -63 \\ \underline{21} \quad \underline{21} \\ n = -3 \end{array} $	$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-8} \qquad \qquad \underline{-8} \\ 4\frac{7}{8} + n = 3 + 4n + 3n \\ 4\frac{7}{8} + n = 3 + 7n \\ \underline{-3} \qquad \underline{-3} \\ 1\frac{7}{8} + n = 7n \\ \underline{-n} \quad \underline{-n} \\ 1\frac{7}{8} = 6n \\ \underline{6} \quad \underline{6} \\ \frac{5}{16} = n \end{array} $

2. Describe the different first steps the students used in solving the equation.
3. Which student made an error?
4. Write a note to the student who made an error explaining the mistake they made.
5. Is -7 a solution to the equation $32 + 7x - 4 = 3x$? Justify your answer mathematically and using words.

For questions 6–10, solve each equation and verify the solution using substitution.

6. $\frac{2m-5}{3} = 4m - 3$

7. $-11 - 2p = 6p + 5(p + 3)$

8. $\frac{2}{5} + 13k - 1 = 5k - \frac{3}{5}$

9. $3.1 + 6.8z - 2.4 = 0.3z - 4$

10. $\frac{1}{2}x - 2(x + 1) = 3 + 1.5x + 1$